Overall survival with sacituzumab govitecan in hormone receptor-positive and human epidermal growth factor receptor 2-negative metastatic breast cancer (TROPiCS-02): a randomised, open-label, multicentre, phase 3 trial

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Summary

Background Sacituzumab govitecan demonstrated significant progression-free survival benefit over chemotherapy in the phase 3 TROPiCS-02 trial in patients with pretreated, endocrine-resistant hormone receptor-positive, human epidermal growth factor receptor 2-negative (HR+ and HER2–) metastatic breast cancer with limited treatment options. Here, we report the protocol-specified final analysis of overall survival and endpoints by trophoblast cell-surface antigen 2 (Trop-2) expression and other variables.

Methods In this randomised, open-label, multicentre, phase 3 trial, which took place in 91 centres across North America (the USA and Canada) and Europe (Belgium, France, Germany, Italy, the Netherlands, Spain, and the UK), patients were randomly assigned (1:1) to receive sacituzumab govitecan or chemotherapy (eribulin, vinorelbine, capecitabine, or gemcitabine). Patients had confirmed HR+ and HER2– locally recurrent inoperable or metastatic breast cancer and had received at least one previous endocrine therapy, a taxane, and a CDK4/6 inhibitor in any setting and two to four previous chemotherapy regimens for metastatic disease. The primary endpoint was progression-free survival (previously reported and not included in this analysis), and secondary endpoints included overall survival, objective response rate (ORR), and patient-reported outcomes. Overall survival was assessed using stratified log-rank tests and Cox regression. Trop-2 expression was assessed in tumour tissue by immunohistochemistry. In the statistical testing hierarchy, ORR and patient-reported outcomes were tested sequentially if overall survival was significant. This study is registered with ClinicalTrials.gov, NCT03901339.

Findings At the data cutoff date of July 1, 2022, 543 of 776 screened patients were randomly assigned between May 30, 2019, and April 5, 2021, with 272 patients in the sacituzumab govitecan group and 271 patients in the chemotherapy group. With a 12·5-month (IQR 6·4–18·8) median follow-up, 390 deaths occurred among 543 patients. Overall survival was significantly improved with sacituzumab govitecan versus chemotherapy (median 14·4 months [95% CI 13·0–15·7] *vs* 11·2 months [10·1–12·7]; hazard ratio [HR] 0·79, 95% CI 0·65–0·96; p=0·020); survival benefit was consistent across Trop-2 expression-level subgroups. ORR was significantly improved with sacituzumab govitecan compared with chemotherapy (57 [21%] patients *vs* 38 [14%]; odds ratio 1·63 [95% CI 1·03–2·56]; p=0·035), as was time to deterioration of global health status and quality of life (median 4·3 months *vs* 3·0 months; HR 0·75 [0·61–0·92]; p=0·0059) and fatigue (median 2·2 months *vs* 1·4 months; HR 0·73 [0·60–0·89]; p=0·0021). The safety profile of sacituzumab govitecan was consistent with previous studies (including the TROPiCS-02 primary analysis and the ASCENT trial). One fatal adverse event (septic shock caused by neutropenic colitis) was determined to be related to sacituzumab govitecan treatment.

Interpretation Sacituzumab govitecan demonstrated statistically significant and clinically meaningful benefit over chemotherapy, with a 3·2-month median overall survival improvement and a manageable safety profile. These data support sacituzumab govitecan as a new treatment option for patients with pretreated, endocrine-resistant HR+ and HER2– metastatic breast cancer.

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Introduction

Hormone receptor-positive, human epidermal growth factor receptor 2-negative (HR+ and HER2–) breast cancer accounts for approximately 70% of breast cancers.[1,2] International guidelines recommend sequential endocrine therapy combined with targeted agents such as cyclin-dependent kinase (CDK)4/6 inhibitors when appropriate, followed by sequential

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Research in context

Evidence before this study

Our literature review included searches of PubMed, international guidelines, major congresses (eg, American Society of Clinical Oncology and European Society for Medical Oncology), and clinical trial websites for clinical trial reports (published in any language, with no date restrictions) evaluating endocrine-based therapy, antibody–drug conjugates, and chemotherapy for the treatment of patients with hormone receptor-positive, human epidermal growth factor receptor 2-negative (HR+ and HER2–) metastatic breast cancer. The search was done on March 6, 2023, and included the search terms "hormone therapy", "endocrine-based therapy", "cyclin-dependent kinase 4/6 inhibitor", "antibody-drug conjugate", "hormone receptor-positive", and "metastatic breast cancer". The current standard of care for the treatment of patients with HR+ and HER2– metastatic breast cancer includes endocrine-based therapy, which can include cyclin-dependent kinase (CDK) 4 and CDK 6 inhibitors and other sequenced targeted agents. However, endocrine resistance eventually develops, after which treatment options have been limited to sequential chemotherapy, often associated with low response rates and cumulative toxicity. Sacituzumab govitecan is a first-in-class antibody–drug conjugate with an SN-38 payload (the active metabolite of the topoisomerase inhibitor irinotecan) targeted to trophoblast cell-surface antigen 2 (Trop-2), a transmembrane calcium signal transducer highly expressed in breast cancer, linked to poor prognosis. Sacituzumab govitecan is approved in several countries for the treatment of patients with unresectable locally advanced or metastatic triple-negative breast cancer who have received at least two previous systemic therapies (at least one for metastatic disease). Sacituzumab govitecan was initially evaluated in HR+ and HER2– metastatic breast cancer in the phase 1–2 single-arm IMMU-132-01 study (NCT01631552) and showed encouraging activity with a safety profile consistent with previous reports. With these results and after review of the literature, discussion with clinicians, researchers, and regulatory agencies, the randomised, phase 3 TROPiCS-02 study (NCT03901339) was initiated. Per protocol, eligible patients must have received at least one line of taxane chemotherapy, at least one anticancer hormonal treatment, and at least one CDK4/6 inhibitor. At the primary analysis of the TROPiCS-02 study, a statistically significant progression-free survival benefit with sacituzumab govitecan over single-agent chemotherapy was shown in patients with pretreated, endocrine HR+ and HER2– metastatic breast cancer.

Added value of this study

At the time of the TROPiCS-02 primary progression-free survival analysis, overall survival was immature and other secondary endpoints were not formally tested on the basis of the study statistical plan. At this planned final analysis, overall survival benefit was observed in patients treated with sacituzumab govitecan versus chemotherapy with no new safety signals. Importantly, based on the prespecified statistical testing hierarchy, the significant overall survival allowed a formal statistical analysis of secondary endpoints, including objective response rate, time to deterioration for global health status and quality of life, and time to deterioration for fatigue, which were all significantly improved with sacituzumab govitecan versus chemotherapy, reinforcing the progression-free survival benefit observed at the primary analysis. Overall survival benefit was observed regardless of tumour Trop-2 expression levels. Based on these analyses, sacituzumab govitecan was approved in the USA in February, 2023, and in the EU in July, 2023, for patients with unresectable locally advanced or metastatic HR+ and HER2– (IHC0, IHC1 positive, or IHC2 positive and ISH negative) breast cancer who have received endocrine- based therapy and at least two additional systemic therapies in the metastatic setting.

Implications of all the available evidence

The planned final overall survival analysis of the phase 3 TROPiCS-02 study demonstrates that sacituzumab govitecan is an effective and safe treatment for patients with pretreated, endocrine-resistant HR+ and HER2– metastatic breast cancer, providing a new therapeutic choice for patients with limited treatment options.

single-agent chemotherapy.[3,4] Later-line chemotherapy is associated with low response rates, poor quality of life, and substantial toxicity.[5–7] There remains an unmet clinical need for novel, effective therapies for endocrine-resistant HR+ and HER2– metastatic breast cancer.

Sacituzumab govitecan is an antibody–drug conjugate composed of a humanised anti-trophoblast cell-surface antigen 2 (Trop-2) monoclonal antibody coupled to SN-38, the active metabolite of the topoisomerase inhibitor irinotecan, via a proprietary hydrolysable linker.[8] Trop-2 is a transmembrane calcium signal transducer linked to tumour progression and poor prognosis and is highly expressed in approximately 80% of breast cancers, regardless of subtype.[9,10] Sacituzumab govitecan is distinguished by high specificity for Trop-2, high drug-to-antibody ratio (7·6:1), a hydrolysable linker facilitating rapid internalisation, and efficient release of payload in cancer cells and in the tumour microenvironment (bystander effect), potentially leading to antitumour effects in adjacent non-Trop-2 expressing cells.[8,11]

Sacituzumab govitecan is approved in several countries for the treatment of patients with unresectable locally advanced or metastatic triple-negative breast cancer who have received at least two previous systemic therapies (at least one for metastatic disease).[12] The randomised, phase 3 TROPiCS-02 trial evaluated sacituzumab govitecan in patients with endocrine-resistant and chemotherapy-resistant HR+ and HER2– metastatic breast cancer. In the primary analysis, sacituzumab

govitecan demonstrated statistically significant improvement in progression-free survival (PFS) compared with single-agent chemotherapy, with a 34% reduction in the risk of disease progression or death (median 5·5 months *vs* 4·0 months; hazard ratio [HR] 0·66, 95% CI 0·53–0·83; p=0·0003).[13] A higher proportion of patients in the sacituzumab govitecan group were alive and were progression free at all landmark timepoints versus those in the chemotherapy group, with three times as many patients being progression free at the 1-year mark (21% *vs* 7%). Sacituzumab govitecan demonstrated a manageable safety profile consistent with that in previous studies.[13,14] At the time of the first prespecified exploratory overall survival analysis, the median overall survival was not yet mature (HR 0·84, p=0·13).[13]

Here, we present results from the final analysis of overall survival from the TROPiCS-02 study of sacituzumab govitecan versus chemotherapy in patients with HR+ and HER2– metastatic breast cancer, including updated, formally tested response and patient-reported outcomes, efficacy by Trop-2 expression, and safety.

Methods

Study design and patients

Details of the TROPiCS-02 trial, including access to the protocol, have been previously reported.[13] Briefly, TROPiCS-02 was a global, randomised, open-label, multicentre, phase 3 study of sacituzumab govitecan versus treatment of physician's choice in patients with locally recurrent inoperable or metastatic HR+ and HER2– breast cancer. TROPiCS-02 enrolled patients at 91 centres across North America (the USA and Canada) and Europe (Belgium, France, Germany, Italy, the Netherlands, Spain, and the UK). Eligible patients had confirmed HR+ and HER2– locally recurrent inoperable or metastatic breast cancer and received at least one previous endocrine therapy, a taxane, and a CDK4/6 inhibitor in any setting and two to four previous chemotherapy regimens for metastatic disease. Other eligibility criteria required patients to be 18 years or older, have an Eastern Cooperative Oncology Group performance status of 0 or 1, and to have at least one measurable target lesion per Response Evaluation Criteria in Solid Tumours (RECIST version 1.1). Key exclusion criteria included previous topoisomerase 1 inhibitor therapy, treatment with chemotherapy, radiation, or small molecule-targeted therapy within 2 weeks and biological therapy within 4 weeks before the first dose of study treatment, active hepatitis B or C infection or active infection requiring antibiotics, history of substantial cardiovascular disease, active chronic inflammatory bowel disease, and history of bowel obstruction.

The study was approved by national regulatory authorities and each investigational site's institutional review or ethics committee before implementation and was compliant with the Declaration of Helsinki and International Council for Harmonisation Good Clinical Practice Guidelines. All patients provided written informed consent.

Randomisation and masking

Patients were randomly assigned 1:1 to receive sacituzumab govitecan (Trodelvy; Gilead Sciences, Foster City, CA, USA) at a dose of 10 mg/kg intravenously on day 1 and day 8 every 21 days or single-agent chemotherapy of physician's choice (eribulin, capecitabine, gemcitabine, or vinorelbine). The chemotherapy agents included in this study, and their recommended doses, were in accordance with locally approved prescribing information or according to National Comprehensive Cancer Network Clinical Practice Guidelines in Oncology for Breast Cancer.[3] Randomisation was stratified by baseline visceral metastases (yes or no), previous endocrine therapy in the metastatic setting for 6 months or longer (yes or no), and number of previous chemotherapy regimens for metastatic disease (two *vs* three to four). Patients received study treatment until disease progression, unacceptable toxicity, withdrawal of consent, or investigator's decision. As this was an open-label trial, participants and people giving the interventions were not masked to group assignment.

Outcomes

The primary endpoint was PFS by masked independent central review per RECIST version 1.1.[15] Further PFS data are not reported in this analysis because the final planned PFS analysis has been completed and reported.[13] Overall survival, objective response rate (ORR), and patient-reported outcomes were key secondary endpoints. The primary and secondary endpoints were updated from the original trial design (described under Major amendments in the appendix p 5). Additional endpoints included clinical benefit rate, duration of response, and safety. Patient-reported outcomes were assessed using the European Organization for Research and Treatment of Cancer Quality of Life Questionnaire (EORTC QLQ-C30). Most efficacy outcomes were assessed in the intention-to-treat population including all randomly assigned patients; patient-reported outcomes were assessed in the EORTC QLQ-C30-evaluable population, defined as patients in the intent-to-treat population with an evaluable EORTC QLQ-C30 assessment at baseline and at least one assessment after baseline. Safety was assessed in patients who received at least one dose of study treatment; adverse event severity was graded per National Cancer Institute Common Terminology Criteria for Adverse Events version 5.0.

See Online for appendix

Procedures

Primary or metastatic archival formalin-fixed paraffin-embedded tissue were requested at study entry to determine tumour Trop-2 expression for correlative analysis. Trop-2 expression was not required to establish

patient eligibility. The Trop-2 immunohistochemistry assay was developed and validated at Roche Tissue Diagnostics (Tucson, AZ, USA) in their College of American Pathologists-certified and Clinical Laboratory Improvement Amendments-certified laboratory using a rabbit monoclonal anti-Trop-2 antibody (clone SP295; Spring Bioscience, Pleasanton, CA, USA). Membrane Trop-2 expression was calculated as a histochemical score (H-score; range, 0–300). Since there was no validated scoring algorithm for Trop-2 immunohistochemistry (IHC) established in breast cancer, H-score was independently scored as a raw number and not according to a prespecified cutoff. Exploratory efficacy analyses were done for H-score subgroups lower than 100 and 100 or higher, resulting in approximately an equal distribution of patients in both subgroups. An additional efficacy analysis was done for patients with a Trop-2 H-score of 10 or less. At baseline, a single whole-blood sample was collected from all patients receiving sacituzumab govitecan for determination of uridine diphosphate glucuronosyltransferase 1A1 (*UGT1A1*) genotype for retrospective assessment of safety. *UGT1A1* genotype was determined by Sanger sequencing (platform ABI 3500XL Genetic Analyzer; Applied Biosystems, Waltham, MA, USA; software ABI Data Collection Software; ABI Sequence Analysis Software, Waltham, MA, USA), done centrally (Covance Genomics Lab; LabCorp, Burlington, NC, USA).

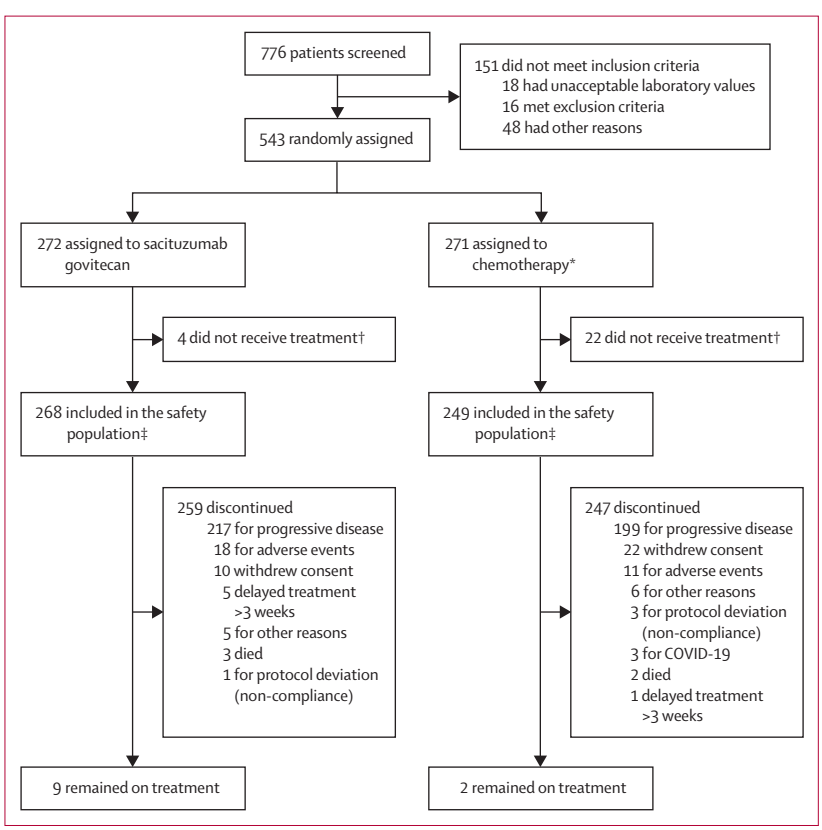

Figure 1: **Trial profile**
*Patients in the chemotherapy group were randomly assigned to eribulin (n=130), vinorelbine (n=63), gemcitabine (n=56), or capecitabine (n=22). †Reasons for patients being randomly assigned but not treated were that one withdrew consent and three withdrew for other reasons in the sacituzumab govitecan group, and 15 withdrew consent, six withdrew for other reasons, and one was withdrawn by sponsor decision in the chemotherapy group. ‡All patients who received at least one dose of study treatment.

Statistical analysis

Study enrolment targeted 520 patients based on statistical considerations for the primary endpoint and key secondary endpoints.[13] Overall survival was described using Kaplan-Meier curves, with median values derived by Kaplan-Meier estimates and associated CIs determined by the Brookmeyer and Crowley method with log-log transformation.[16] Based on the prespecified statistical testing hierarchy (appendix p 6), overall survival would be formally tested if PFS was statistically significant. For the planned final analysis of overall survival, 350 events were targeted, with 390 events observed; testing was done at the two-sided significance level of 0·0223, adjusted for the actual number of events. The Lan DeMets alpha spending function approximating a Pocock approach accounted for multiplicity. ORR was formally tested if overall survival was statistically significant, followed by patient-reported outcomes, if each prior endpoint in the hierarchy was significant. The graphical approach of Maurer and Bretz was used to control multiplicity for patient-reported outcomes.[17] All statistical analyses were done using SAS, version 9.4. Additional methods, including rationale for choice of alpha spending strategy for overall survival and other major amendments, are outlined in the appendix (p 5).

Role of the funding source

The funder of the study was involved in study design, data collection, data analysis, data interpretation, and writing of this report.

Results

At the data cutoff date of July 1, 2022, 543 of 776 screened patients were randomly assigned between May 30, 2019, and April 5, 2021, with 272 patients in the sacituzumab govitecan group and 271 patients in the chemotherapy group (figure 1). In the sacituzumab govitecan group 268 (99%) received study treatment and in the chemotherapy group 249 (92%) patients received study treatment. Within the chemotherapy group, 130 (48%) patients received eribulin, 63 (23%) vinorelbine, 56 (21%) gemcitabine, and 22 (8%) capecitabine. At the data cutoff date, nine (3%) patients in the sacituzumab govitecan group and two (1%) patients in the chemotherapy group remained on treatment. Most patients discontinued treatment because of progressive disease.

Patient demographics and characteristics were generally well balanced across treatment groups as previously reported (table 1).[13] The representativeness of

	Sacituzumab govitecan (n=272)	Chemotherapy (n=271)
Sex		
Female	270 (99%)	268 (99%)
Male	2 (1%)	3 (1%)
Age in years, median (IQR)	57 (49–65)	55 (48–63)
Visceral metastases at baseline		
Yes	259 (95%)	258 (95%)
No*	13 (5%)	13 (5%)
Previous CDK4/6 inhibitor use		
≤12 months	161 (59%)	166 (61%)
>12 months	106 (39%)	102 (38%)
Unknown	5 (2%)	3 (1%)
Initial line of therapy for use of CDK4/6 inhibitor in the metastatic setting		
First line	84 (31%)	101 (37%)
Second line	57 (21%)	56 (21%)
Third line	55 (20%)	49 (18%)
Fourth line or more	71 (26%)	62 (23%)
Previous PARP inhibitor use		
Olaparib	8 (3%)	13 (5%)
Talazoparib†	8 (3%)	5 (2%)
Niraparib	0	1 (<1%)
Rucaparib	1 (<1%)	0
Previous endocrine therapy in the metastatic setting for more than 6 months		
Yes	235 (86%)	234 (86%)
No	37 (14%)	37 (14%)
Previous chemotherapy regimens in the metastatic setting, median (IQR)‡	3 (2–3)	3 (2–3)
Previous chemotherapy regimens in the metastatic setting‡		
None	1 (<1%)	0
One	8 (3%)	2 (1%)
Two	104 (38%)	118 (43%)
Three or more	159 (58%)	151 (56%)
Evaluable Trop-2 expression	238 (88%)	224 (83%)
H-score <100§	96 (40%)	96 (43%)
H-score ≥100§	142 (60%)	128 (57%)

Data are n (%) unless otherwise stated. Additional patient demographics and baseline characteristics have been published previously.[33] H-score=histochemical score. PARP=poly (ADP-ribose) polymerase. Trop-2=trophoblast cell-surface antigen 2. *Generally includes metastases in soft tissue involving the liver, lungs, adrenal glands, peritoneum, pleura, and brain. †Talazoparib includes the formulation talazoparib tosylate. ‡The reported number of previous therapies were miscounted at screening for some patients. Nine patients had fewer or more previous chemotherapy regimens in the metastatic setting than the specified inclusion criteria and were included in the intent-to-treat population. §Percentages for this subgroup refer to the proportion of patients with available Trop-2 data. CDK=cyclin-dependent kinase.

Table 1: **Patient demographics and baseline characteristics**

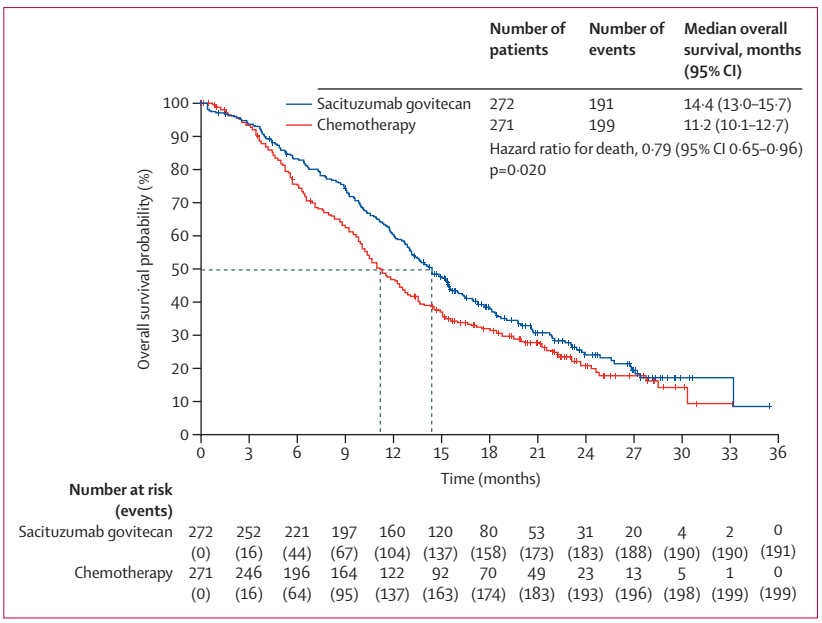

Figure 2: **Overall survival in the intention-to-treat population**
This is the planned final analysis of overall survival in the intent-to-treat population of all randomly assigned patients. Vertical hash marks indicate censored data.

study participants is shown in the appendix (p 9). Patients had a median age of 56 years (IQR 49–65), and median time from initial metastatic diagnosis to randomisation in the study was 47·8 months (IQR 31·2–67·7). Patients had received a median of three previous lines of chemotherapy (IQR 2–3) for advanced disease. Most patients received previous endocrine therapy in the metastatic setting for 6 months or more (469 [86%]). CDK4/6 inhibition in combination with endocrine therapy was given to 84 (31%) patients in the sacituzumab govitecan group and 101 (37%) patients in the control group in the first-line endocrine setting, and 183 (67%) in the sacituzumab govitecan group and 167 (62%) in the control group in the second-line or later-line setting; 208 (38%) received previous CDK4/6 inhibitor therapy for more than 12 months. Demographics for patients with one line or two or more lines of previous CDK4/6 inhibitor therapy are included in the appendix (p 11). In total, 462 (85%) patients had available Trop-2 IHC data (samples collected at a median of 7·7 months [IQR 0·5–24·7] before randomisation); Trop-2 expression was observed in 437 (95%) patients with evaluable samples. In patients with Trop-2 data, 192 (42%) had H-scores lower than 100 and 270 (58%) had H-scores of 100 or higher. Baseline demographics for patients with Trop-2 H-scores lower than 100 and 100 or higher were generally similar and consistent with that of the overall study population (appendix p 12).

With a median duration of follow-up of 12·5 months (IQR 6·4–18·8; 13·8 months [8·3–19·8] with sacituzumab govitecan and 10·7 months [5·6–18·3] with chemotherapy), there were 191 overall survival events in the sacituzumab govitecan group and 199 events in the chemotherapy group. Treatment with sacituzumab govitecan resulted in a significant overall survival benefit compared with chemotherapy (figure 2), representing a median 3·2-month improvement. In a landmark analysis, the 12-month overall survival rate was 61%

	Sacituzumab govitecan (n=272)	Chemotherapy (n=271)
Overall number of survival events	191	199
Overall survival in months, median (95% CI)	14·4 (13·0–15·7)	11·2 (10·1–12·7)
HR (95% CI), p=0·020	0·79 (0·65–0·96)	
Overall survival rate, % (95% CI)		
12 months	61% (55–66)	47% (41–53)
18 months	39% (33–45)	32% (27–38)
24 months	25% (19–31)	21% (16–27)
ORR, n (%)*	57 (21%)	38 (14%)
OR (95% CI), p=0·035	1·63 (1·03–2·56)	
Best overall response, n (%)*		
Complete response	2 (1%)	0
Partial response	55 (20%)	38 (14%)
Stable disease	142 (52%)	106 (39%)
Stable disease for 6 months or longer	35 (13%)	22 (8%)
Progressive disease	58 (21%)	76 (28%)
Not evaluable	15 (6%)	51 (19%)
Clinical benefit rate, n (%)†	92 (34%)	60 (22%)
OR (95% CI), nominal p=0·0025	1·80 (1·23–2·63)	
Duration of response in months, median (95% CI)	8·1 (6·7–9·1)	5·6 (3·8–7·9)

On the basis of the statistical testing hierarchy, ORR was tested sequentially for significance given that overall survival was significant. ORR was not formally tested previously.[33] HR=hazard ratio. OR=odds ratio. ORR=objective response rate. *Assessed by masked independent central review per the Response Evaluation Criteria in Solid Tumours version 1.1. †Clinical benefit rate is defined as the proportion of patients with a confirmed best overall response of complete response, partial response, and stable disease for 6 months or longer.

Table 2: **Summary of treatment efficacy**

(95% CI 55–66) for the sacituzumab govitecan group and 47% (41–53) in the chemotherapy group; the 18-month and 24-month rates were also numerically higher with sacituzumab govitecan versus chemotherapy (table 2).

Overall survival benefit for sacituzumab govitecan versus chemotherapy was generally consistent across predefined subgroups, including patients who received three or more previous chemotherapy regimens in the metastatic setting, those with visceral metastases, and those who received endocrine therapy in the metastatic setting for 6 months or longer (figure 3). Patients had an overall survival benefit regardless of age. Although more patients received CDK4/6 inhibition in the second-line or in later-line settings, overall survival benefit was seen regardless of line of use (median 13·1 *vs* 10·2 months for first-line therapy and median 15·3 months *vs* 12·1 months in the second-line or later-line therapies; appendix p 13). Overall survival benefit for sacituzumab govitecan over chemotherapy was also seen regardless of Trop-2 expression (H-score <100, median 14·6 months *vs* 11·3 months; H-score ≥100, median 14·4 months *vs* 11·2 months; appendix p 14), including in 79 patients (sacituzumab govitecan n=34; chemotherapy n=45) with low Trop-2 expression (H-score ≤10, median 17·6 months *vs* 12·3 months; HR 0·61, 95% CI 0·34–1·08).

The ORR by masked independent central review was significantly higher with sacituzumab govitecan versus chemotherapy (table 2); 55 (20%) patients in the sacituzumab govitecan group and 38 (14%) in the chemotherapy group reached a partial response. Clinical benefit was higher with sacituzumab govitecan versus chemotherapy. Median duration of response was 8·1 months (95% CI 6·7–9·1) for sacituzumab govitecan versus 5·6 months (3·8–7·9) for chemotherapy.

Median time to deterioration for global health status or quality of life (4·3 months *vs* 3·0 months; HR 0·75, 95% CI 0·61–0·92; p=0·0059; appendix p 7) and fatigue (2·2 months *vs* 1·4 months; HR 0·73, 0·60–0·89; p=0·0021; appendix p 8) was significantly longer with sacituzumab govitecan versus chemotherapy, whereas median time to deterioration for pain was similar between treatment groups (3·8 months *vs* 3·5 months; HR 0·92, 0·75–1·13; p=0·42).

In patients who received at least one dose of study treatment, median treatment duration was 4·1 months (IQR 1·2–8·0) for sacituzumab govitecan and 2·3 months (1·0–5·1) for chemotherapy; median relative dose intensity with sacituzumab govitecan was 98·9%, compared with 96·4% for eribulin, 90·8% for capecitabine, 86·4% for gemcitabine, and 98·0% for vinorelbine. A summary of treatment-emergent adverse events and common treatment-related adverse events is shown (table 3; appendix p 15). The most frequent treatment-emergent adverse events of any grade with sacituzumab govitecan versus chemotherapy were neutropenia (189 [71%] *vs* 136 [55%]), diarrhoea (166 [62%] *vs* 57 [23%]), nausea (157 [59%] *vs* 87 [35%]), alopecia (128 [48%] *vs* 46 [18%]), and anaemia (98 [37%] *vs* 69 [28%]); the most common grade 3 or worse treatment-emergent adverse events were neutropenia (138 [51%] *vs* 97 [39%]) and diarrhoea (27 [10%] *vs* 3 [1%]). These incidence rates of the most common treatment-emergent adverse events in this updated analysis were consistent with the previous study report.[13] The number of patients in the study population who initiated granulocyte colony-stimulating factor at any time during the study was 145 (54%) in the sacituzumab govitecan group and 85 (34%) in the chemotherapy group as previously reported.[13]

Grade 3 or worse treatment-related febrile neutropenia occurred in 14 (5%) patients in the sacituzumab govitecan group and 11 (4%) patients in the chemotherapy group. Treatment-related all-grade neuropathy occurred in 24 (9%) patients in the sacituzumab govitecan group versus 39 (16%) in the chemotherapy group (appendix p 15). There were no cases of treatment-related pneumonitis in the sacituzumab govitecan group; in the chemotherapy group, two patients had treatment-related all-grade pneumonitis (one case of grade 3 or worse pneumonitis).

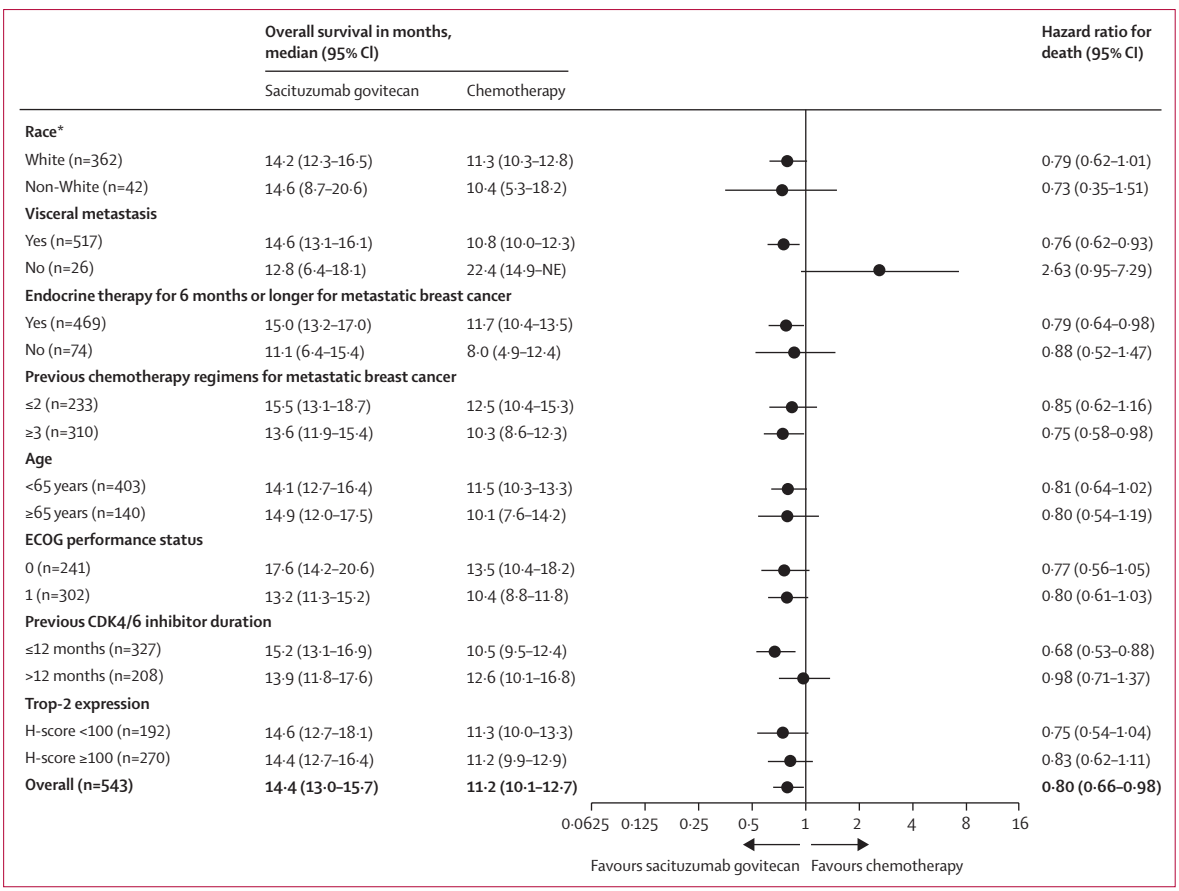

Figure 3: **Subgroup analysis of overall survival**

Hazard ratios are from unstratified Cox regression, including overall and subgroup hazard ratios. Subgroup analyses of overall survival by race, geographical region, chemotherapy group agents, and early relapse were assessed but not included in this figure. CDK4/6=cyclin-dependent kinase 4 and 6. ECOG=Eastern Cooperative Oncology Group. H-score=histochemical score. NE=not evaluable. Trop-2=trophoblast cell-surface antigen 2. *In some regions, local regulators did not allow collection of race or ethnicity data. There were 69 patients in the sacituzumab govitecan group and 70 patients in the chemotherapy group who did not have race or ethnicity specified, and these patients were not included in the race subgroup analysis.

Adverse events leading to treatment discontinuation were infrequent, occurring in 17 (6%) patients in the sacituzumab govitecan group and 11 (4%) patients in the chemotherapy group. In the sacituzumab govitecan group, six patients had fatal adverse events; one was determined to be treatment related (septic shock caused by neutropenic colitis), and no pattern was identified in the remaining fatal adverse events, which included COVID-19 pneumonia, pulmonary embolism, pneumonia, nervous system disorder, and arrhythmia. No fatal adverse events were reported in the chemotherapy group.

Of patients treated with sacituzumab govitecan in TROPiCS-02, 103 (38%) had wild-type, 119 (44%) had heterozygous, and 25 (9%) had homozygous *UGT1A1* genotypes. Patients who were homozygous for the *UGT1A1* *28/*28 genotype had numerically higher rates of grade 3 or worse treatment-emergent adverse events and adverse events leading to discontinuation than patients with wild-type (*1/*1) or heterozygous (*1/*28) *UGT1A1* genotypes (appendix p 17). Patients who were homozygous for *UGT1A1* *28/*28 also had higher rates of any-grade anaemia and grade 3 or worse diarrhoea and neutropenia than the other subgroups, although sample sizes were small in the homozygous subgroup (appendix p 18).

Discussion

In this planned final overall survival analysis from the TROPiCS-02 study in pretreated, endocrine-resistant HR+ and HER2– metastatic breast cancer, sacituzumab govitecan demonstrated a statistically significant improvement in overall survival compared with single-agent chemotherapy. Sacituzumab govitecan treatment resulted in a 21% relative reduction in risk of death, and the corresponding 3·2-month improvement in median overall survival compared with chemotherapy is clinically meaningful in the metastatic setting. At both the primary analysis and the planned final analysis of this study, the adverse events reported remained consistent with the known safety profile of sacituzumab govitecan.[13,14] In this study, Trop-2 expression was observed in 437 (95%) patients with evaluable samples. Similar to

	Sacituzumab govitecan (n=268)	Chemotherapy (n=249)
Grade 3 or higher	198 (74%)	150 (60%)
Leading to treatment discontinuation	17 (6%)	11 (4%)
Leading to dose delay	178 (66%)	109 (44%)
Leading to dose reduction	90 (34%)	82 (33%)
Serious events	74 (28%)	48 (19%)
Leading to death*	6 (2%)	0
Treatment-related death	1 (<1%)	0

Treatment-emergent adverse events were defined as any adverse event that began or worsened on or after the start of the study drug until 30 days after the last dose of the study drug. *Of six treatment-emergent adverse events leading to death, only one was considered by the investigator to be treatment related (septic shock caused by neutropenic colitis). The other five deaths were caused by COVID-19 pneumonia, pulmonary embolism, pneumonia, nervous system disorder, and arrhythmia. Upon detailed review of the treatment-emergent adverse events leading to death, no patterns were identified.

Table 3: **Summary of treatment-emergent adverse events**

biomarker analyses from the phase 3 ASCENT study in pretreated metastatic triple-negative breast cancer,[18] this post-hoc analysis demonstrated that sacituzumab govitecan improves outcomes in patients with HR+ and HER2– metastatic breast cancer compared with treatment of physician's choice regardless of Trop-2 expression levels. The high affinity antibody and the anti-tumour bystander effect might contribute to this finding as outlined hereafter.

Sacituzumab govitecan is a Trop-2-directed antibody–drug conjugate distinguished by its high specificity and nanomolar affinity for Trop-2. The antibody binds to Trop-2 expressed on the surface of tumour cells, allowing for targeted delivery of the payload (SN-38) to tumour cells. SN-38 is also released in the tumour microenvironment without internalisation or enzymatic cleavage of SN-38 from the antibody, leading to an anti-tumour bystander effect regardless of Trop-2 expression. On the basis of these data, Trop-2 expression testing before treatment with sacituzumab govitecan is not beneficial in determining drug efficacy.

Sacituzumab govitecan also demonstrated significant improvement in ORR and significantly delayed deterioration in fatigue and global health status and quality of life outcomes compared with chemotherapy. The safety profile was consistent with previous reports;[13,14,19] no new safety signals emerged with longer follow-up. Treatment with sacituzumab govitecan in this trial resulted in no cases of pneumonitis, a key adverse event of concern with other classes of antibody–drug conjugates.[20] Neutropenia and diarrhoea, known adverse events associated with sacituzumab govitecan, are generally adequately managed with established supportive-care measures; one death caused by septic shock secondary to neutropenic colitis was observed.[13,14] Polymorphisms in *UGT1A1* have been associated with increased incidence of known adverse events, such as neutropenia, febrile neutropenia, anaemia, and diarrhoea, following treatment with some systemic anticancer agents, including sacituzumab govitecan, because of the reduced rate of SN-38 glucuronidation.[21] Patients who were homozygous for *UGT1A1* *28/*28 had numerically higher rates of grade 3 or worse treatment-emergent adverse events, treatment-emergent adverse events leading to discontinuation, any-grade anaemia, and grade 3 or worse diarrhoea and neutropenia, although the sample size was small, limiting interpretation of these analyses. In the ASCENT trial, homozygosity for *UGT1A1* was associated with a modest increase in risk of neutropenia and diarrhoea, which were managed expectantly.[21] The impact of *UGT1A1* polymorphisms on patient safety will continue to be evaluated in ongoing trials. At present, *UGT1A1* testing is not recommended before sacituzumab govitecan treatment.

The overall survival benefit for sacituzumab govitecan versus chemotherapy was generally consistent across predefined subgroups. CDK4/6 inhibitors were given more frequently in the second or greater endocrine line setting, with similar effects on overall survival regardless of sequencing of CDK4/6 inhibitor use. Of note, patients who received CDK4/6 inhibitors for 12 months or less had a larger numerical overall survival benefit (15·2 months *vs* 10·5 months) than those who had received CDK4/6 inhibitors for more than 12 months (13·9 months *vs* 12·6 months). These results might be affected by sequencing of CDK4/6 inhibitor use and could be explained by studies demonstrating poorer response to therapy following CDK4/6 inhibitor use.[20,22,23] It is difficult to evaluate the overall survival benefit of sacituzumab govitecan in patients without baseline visceral metastases because of the very small sample size (n=26 across both groups). However, a significant PFS benefit with sacituzumab govitecan was noted in this subset. Given these small numbers, the generalisability of additional clinical outcomes to patients without baseline visceral metastases is limited.

Treatment for patients with endocrine-resistant HR+ and HER2– metastatic breast cancer who have received several lines of chemotherapy represents an unmet need. All patients enrolled in TROPiCS-02 had received guideline-directed therapy, including endocrine therapy, a CDK4/6 inhibitor, a taxane, and a median of three lines (IQR 2–3) of chemotherapy for advanced disease.[3] Before the development of CDK4/6 inhibitors, the most recent previous late-line study compared eribulin with treatment of choice for metastatic breast cancer (Study 301) and demonstrated improved overall survival with eribulin in a heterogeneous patient population.[6] However, in a pooled subgroup analysis of the two phase 3 trials EMBRACE and Study 301, overall survival was not significantly improved with eribulin in patients with HR+ disease.[24] By comparison, sacituzumab govitecan

demonstrated both improved PFS and overall survival over chemotherapy in the endocrine-resistant, post-CDK4/6 inhibitor setting.

Patients with HER2 IHC0 disease, excluded from the phase 3 DESTINY-Breast04 study, represent at least a third of HR+ breast cancers. DESTINY-Breast04 reported improved PFS and overall survival with trastuzumab deruxtecan compared with single-agent chemotherapy in patients with HR+ and HER2-low (HER2 IHC1 positive or IHC2 positive and in-situ-hybridisation [ISH] negative) metastatic breast cancer, and in the overall study population, which included 58 patients with HR-negative and HER2-low disease.[20] With a median of one previous line of chemotherapy for metastatic disease and approximately 70% having received previous CDK4/6 inhibitors, the HR+ and HER2-low DESTINY-Breast04 population was less heavily pretreated than the TROPiCS-02 population.[20] Guidance from the National Comprehensive Cancer Network specifies that sacituzumab govitecan can be used in the second-line setting for unresectable locally advanced or metastatic HR+ and HER2– breast cancer if the patient is not a candidate for trastuzumab deruxtecan as a category one preferred regimen, and can be considered for treatment beyond the second-line setting.[3] Sacituzumab govitecan is also being evaluated in earlier settings in patients with HR+ and HER2– metastatic breast cancer who have received endocrine therapy in the ASCENT-07 study (NCT05840211).

Sacituzumab govitecan and trastuzumab deruxtecan have different antibody targets and linkers, and their payloads could be non-cross resistant despite targeting the same enzyme. Resistance mechanisms to antibody–drug conjugates are not well defined but might involve several mechanisms such as downregulation or altered expression of target biomarkers, alterations in lysosomal activity or drug efflux pumps, or alternative signalling pathway activation.[25,26] Previous post-hoc analyses indicate that sacituzumab govitecan is an effective option for patients with HR+ and HER2– breast cancer regardless of HER2 IHC0 or HER2-low status.[27–29] Additional studies to understand resistance mechanisms will help guide therapeutic sequencing strategies for these novel agents and studies are being planned to help understand the efficacy of sequencing various antibody–drug conjugates in the metastatic setting.

Potential limitations of this study have been previously described.[13] Briefly, 22 (8%) patients randomly assigned to the chemotherapy group were not treated. Additionally, most patients had visceral metastases (517 [95%]) and were heavily pretreated with chemotherapy, all factors associated with reduced efficacy benefit and higher risk of neutropenia, and masked independent central review assessment of progressive disease was not done in real time, which might have increased patient censoring. Differences in previous treatments and in physician's choice of chemotherapy in the study might have influenced efficacy. Lastly, hormone receptor status was assessed locally at any stage of disease.

TROPiCS-02 represents the largest clinical trial to date specifically done in patients with endocrine-resistant and chemotherapy-resistant HR+ and HER2– metastatic breast cancer who have received previous endocrine and CDK4/6 inhibitor therapy, reflecting current standard of care. This updated efficacy analysis from TROPiCS-02 demonstrated a statistically significant overall survival benefit for sacituzumab govitecan over chemotherapy, supporting the previously reported PFS data.[13] Combined with delayed deterioration in quality of life and a manageable safety profile consistent with previous reports,[13] these efficacy findings indicate that sacituzumab govitecan is an important novel therapy for these patients. On the basis of this prespecified formal overall survival analysis of the TROPiCS-02 study presented herein, sacituzumab govitecan was approved in the USA and in the EU for patients with unresectable locally advanced or metastatic HR+ and HER2– (IHC0, IHC1 positive, or IHC2 positive and ISH negative) breast cancer who have received endocrine-based therapy and at least two additional systemic therapies in the metastatic setting. An exploratory, descriptive analysis of overall survival with longer follow-up for this study is planned.

Contributors

All authors were involved in the original drafting and revision of the manuscript for publication, visualisation, methodology, and final approval of the manuscript. HSR, AB, FM, JC, PS, RD, TV, HW, MM, OKY, WV, and SMT were involved in the conceptualisation and design of the study. HSR, AB, FM, JC, DL, OT, EC, FD, PGP, KLJ, and SMT were involved in acquiring resources, the investigation of the study, project administration, supervision, and validation of the study. RD, TV, HW, MM, OKY, and WV were involved in data analysis and all authors were involved in data interpretation. All authors had full access to all the data in the study and provided final approval to submit the manuscript for publication.

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Data sharing

Gilead Sciences shares anonymised individual patient data upon request or as required by law or regulation with qualified external researchers based on submitted curriculum vitae and reflecting no conflict of interest. The request proposal must also include a statistician. Approval of such requests is at Gilead Science's discretion and is dependent on the nature of the request, the merit of the research proposed, the availability of the data, and the intended use of the data. Data requests should be sent to datarequest@gilead.com.

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References

1 Kohler BA, Sherman RL, Howlader N, et al. Annual report to the nation on the status of cancer, 1975-2011, featuring incidence of breast cancer subtypes by race/ethnicity, poverty, and state. *J Natl Cancer Inst* 2015; **107:** djv048.

2 Howlader N, Altekruse SF, Li CI, et al. US incidence of breast cancer subtypes defined by joint hormone receptor and HER2 status. *J Natl Cancer Inst* 2014; **106:** dju055.

3 National Comprehensive Cancer Network. NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines) Breast Cancer, version 4. 2023. https://www.nccn.org/professionals/physician_gls/pdf/breast.pdf (accessed May 12, 2023).

4 Gennari A, André F, Barrios CH, et al. ESMO Clinical Practice Guideline for the diagnosis, staging and treatment of patients with metastatic breast cancer. *Ann Oncol* 2021; **32:** 1475–95.

5 Cortes J, O'Shaughnessy J, Loesch D, et al. Eribulin monotherapy versus treatment of physician's choice in patients with metastatic breast cancer (EMBRACE): a phase 3 open-label randomised study. *Lancet* 2011; **377:** 914–23.

6 Twelves C, Awada A, Cortes J, et al. Subgroup analyses from a phase 3, open-label, randomized study of eribulin mesylate versus capecitabine in pretreated patients with advanced or metastatic breast cancer. *Breast Cancer* 2016; **10:** 77–84.

7 Yuan P, Hu X, Sun T, et al. Eribulin mesilate versus vinorelbine in women with locally recurrent or metastatic breast cancer: a randomised clinical trial. *Eur J Cancer* 2019; **112:** 57–65.

8 Goldenberg DM, Cardillo TM, Govindan SV, Rossi EA, Sharkey RM. Trop-2 is a novel target for solid cancer therapy with sacituzumab govitecan (IMMU-132), an antibody-drug conjugate (ADC). *Oncotarget* 2015; **6:** 22496–512.

9 Ambrogi F, Fornili M, Boracchi P, et al. Trop-2 is a determinant of breast cancer survival. *PLoS One* 2014; **9:** e96993.

10 Vidula N, Yau C, Rugo H. Trophoblast cell surface antigen 2 gene (TACSTD2) expression in primary breast cancer. *Breast Cancer Res Treat* 2022; **194:** 569–75.

11 Kopp A, Hofsess S, Cardillo TM, Govindan SV, Donnell J, Thurber GM. Antibody drug conjugate sacituzumab govitecan drives efficient tissue penetration and rapid intracellular drug release. *Mol Cancer Ther* 2023; **22:** 102–11.

12 Trodelvy. Sacituzumab govitecan hziy. Prescribing information. Foster City, CA, USA: Gilead Sciences, 2022.

13 Rugo HS, Bardia A, Marmé F, et al. Sacituzumab govitecan in hormone receptor-positive/human epidermal growth factor receptor 2-negative metastatic breast cancer. *J Clin Oncol* 2022; **40:** 3365–76.

14 Bardia A, Hurvitz SA, Tolaney SM, et al. Sacituzumab govitecan in metastatic triple-negative breast cancer. *N Engl J Med* 2021; **384:** 1529–41.

15 Eisenhauer EA, Therasse P, Bogaerts J, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer* 2009; **45:** 228–47.

16 Brookmeyer R, Crowley J. A confidence interval for the median survival time. *Biometrics* 1982; **38:** 29–41.

17 Maurer W, Bretz F. Multiple testing in group sequential trials using graphical approaches. *Stat Biopharm Res* 2013; **5:** 311–20.

18 Bardia A, Tolaney SM, Punie K, et al. Biomarker analyses in the phase III ASCENT study of sacituzumab govitecan versus chemotherapy in patients with metastatic triple-negative breast cancer. *Ann Oncol* 2021; **32:** 1148–56.

19 Kalinsky K, Diamond JR, Vahdat LT, et al. Sacituzumab govitecan in previously treated hormone receptor-positive/HER2-negative metastatic breast cancer: final results from a phase I/II, single-arm, basket trial. *Ann Oncol* 2020; **31:** 1709–18.

20 Modi S, Jacot W, Yamashita T, et al. Trastuzumab deruxtecan in previously treated HER2-low advanced breast cancer. *N Engl J Med* 2022; **387:** 9–20.

21 Rugo HS, Tolaney SM, Loirat D, et al. Safety analyses from the phase 3 ASCENT trial of sacituzumab govitecan in metastatic triple-negative breast cancer. *NPJ Breast Cancer* 2022; **8:** 98.

22 Chainitikun S, Long JP, Rodriguez-Bautista R, et al. The efficacy of first-line chemotherapy in endocrine-resistant hormone receptor-positive (HR+), human epidermal growth factor receptor 2-negative (HER2-) metastatic breast cancer. *Breast Cancer Res Treat* 2020; **183:** 729–39.

23 Mo H, Renna CE, Moore HCF, et al. Real-world outcomes of everolimus and exemestane for the treatment of metastatic hormone receptor-positive breast cancer in patients previously treated with CDK4/6 inhibitors. *Clin Breast Cancer* 2022; **22:** 143–48.

24 Twelves C, Cortes J, Vahdat L, et al. Efficacy of eribulin in women with metastatic breast cancer: a pooled analysis of two phase 3 studies. *Breast Cancer Res Treat* 2014; **148:** 553–61.

25 Díaz-Rodríguez E, Gandullo-Sánchez L, Ocaña A, Pandiella A. Novel ADCs and strategies to overcome resistance to anti-HER2 ADCs. *Cancers* 2021; **14:** 154.

26 Coates JT, Sun S, Leshchiner I, et al. Parallel genomic alterations of antigen and payload targets mediate polyclonal acquired clinical resistance to sacituzumab govitecan in triple-negative breast cancer. *Cancer Discov* 2021; **11:** 2436–45.

27 Schmid P, Cortés J, Marmé F, et al. Sacituzumab govitecan efficacy in hormone receptor-positive/human epidermal growth factor receptor 2-negative metastatic breast cancer by HER2 immunohistochemistry status in the phase III TROPiCS-02 study. *Ann Oncol* 2022; **33** (suppl 7)**:** S88–121.

28 Gampenrieder SP, Rinnerthaler G, Tinchon C, et al. Landscape of HER2-low metastatic breast cancer (MBC): results from the Austrian AGMT_MBC-Registry. *Breast Cancer Res* 2021; **23:** 112.

29 Denkert C, Seither F, Schneeweiss A, et al. Clinical and molecular characteristics of HER2-low-positive breast cancer: pooled analysis of individual patient data from four prospective, neoadjuvant clinical trials. *Lancet Oncol* 2021; **22:** 1151–61.